2024 2nd International Conference on Recent Trends in Microelectronics, Automation, Computing and Communications Systems (ICMACC)

2024 2nd International Conference on Recent Trends in Microelectronics, Automation, Computing and Communications Systems (ICMACC) | 979-8-3503-6657-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICMACC62921.2024.10891

Sea Erosion Prediction and Monitoring using IOT and ML

Dr.Usha Bala V
Dept. of Computer Science & Engineering,
Anil Neerukonda Institute of Technology & Sciences
Visakhapatnam,India.
Ushabala.cse@anits.edu.in

Harshavardhan Sai.Y
Dept.of Computer Science &Enginering,
Anil Neerukonda Institute of Technology & Sciences
Visakapatnam,India.

Slipa Shramila V
Dept. of Computer Science & Engineering ,
Anil Neerukonda Institute of Technology & Sciences
Visakhapatnam,India

Asritha M
Dept of Computer Science & Engineering,
Anil Neerukonda Institute of Technology & Sciences
Visakhapatnam,India.

Bharath M
Dept. of Computer Scince & Engineering,
Anil Neerukonda Institute of Technology & Sciences
Visakhapatnam,India

A.Chandu
Dept. of Computer Science & Engineering,
Anil Neerukonda Institute of Technology & Sciences
Visakhapatnam,India.

***Abstract*- One of the major environmental issues that the world is currently facing is sea erosion, which requires an innovative solution for early detection and mitigation. Our paper focuses on the integrated system that combines both IoT and Machine Learning. The four main important environmental factors that are responsible for sea erosion are temperature, wave height, wave speed, and soil moisture. IoT sensors deployed along the prototype design are used to monitor these environmental parameters. Another key component is the machine learning algorithm, which is used analyse the interactions, patterns, and relations among the environmental data. By analysing the patterns among the previous data, the model can predict the likelihood of sea erosion events. This allows for the early detection of erosion events and timely interventions that can minimize the damage or loss caused by sea erosion. The main aim of this project is to help the coastal management authority take smart decisions and proactive steps using both real-time and past data analysis.**

Keywords: Sea erosion, IoT, Machine learning, temperature, wave speed, wave height, soil moisture.

I. INTRODUCTION

1.1 Background

Sea erosion is an increasing threat that is faced worldwide. Sea erosion is the process of wearing away coastal land due to an increase in sea level and strong waves. Sea erosion involves the removal and displacement of rocks and sand from coastal areas. Several methods have been introduced previously, but these traditional methods of monitoring and predicting sea erosion often fall short, necessitating innovative approaches that leverage cutting-edge technologies. This paper introduces an innovative approach by combining IoT devices and machine learning techniques to enhance sea erosion prediction and monitoring. The use of IoT helps to fetch data for analysis and provides a more comprehensive understanding of coastal environments. By using machine learning algorithms, we develop a predictive model capable of learning patterns and forecasting erosion events with higher accuracy.

1.2 Motivation

The goal of this research is to address the growing problems brought by coastal erosion, which endangers ecosystems, infrastructure, human populations, and natural habitats. When it comes to adapting to quickly shifting coastal circumstances, traditional monitoring techniques frequently fall short. To solve this, ML and IoT technologies are being combined to build a dynamic, networked monitoring system that allows real-time data collection from four sensors. Sensors are used to collect data on temperature, soil moisture, wave speed, and height. These sensors are deployed along the prototype design, which resembles the actual coastal areas. Decision-makers may take preventive steps to lessen the consequences of erosion based on the constant stream of data that offers insights in real-time. The use of the machine learning approach allows us to create a model that gives accurate predictions and can adjust to different situations.

1.3 Objectives

Firstly, we aim to design and implement a prototype that replicates a coastal environment, incorporating sand and deploying four key sensors to collect data on temperature [1], soil moisture, wave speed, and height. These sensors are the backbone of our project and are used for the collection of data. These sensors are then integrated with the microcontroller.

Secondly, we collect the data from the sensors that are deployed along the prototype design by generating the waves. This dataset is the foundation for the training and testing of machine learning models.

979-8-3503-6657-0/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:05:32 UTC from IEEE Xplore. Restrictions apply.

The third objective involves the integration of the ML model with the user-friendly web interface. The user interface provides four fields where the user can enter the live sensor readings and predict the occurrence of sea erosion events. The user interface also provides an alert system in case of an of an erosion event.

II. LITERATURE REVIEW

This review of the literature presents an outline of recent studies on the coastal erosion monitoring and prediction. Roberto Girau et al. (2019) [2] used the Internet of Things (IoT) to solve the coastal erosion issue. They presented a "Coastal Monitoring System" based on the social internet of things. They considered beach overcrowding and environmental changes to be the main causes of the coastal erosion. They have used two main components. The Beach Unit (BU), which is installed at the beach, is used for crowd detection, and the Sea Unit (SU), which floats on the sea, is used to measure key environmental parameters like temperature, wave height, motions of waves, etc. The classifier model performs with an accuracy of 94.30% and 87.32% for low and high crowds, respectively. Papakonstantinou et al. [3]: This study employs UAV-SFM technology to detect coastal erosion caused by wind and wave events. The use of computer vision algorithms enables precise 3D representations, revealing a significant 6-meter coastline shift after an extreme event. Georeferenced models derived from low-altitude UAV flights offer valuable insights for coastal management and risk assessment.

A case study of a micro-tidal sandy beach in Tamil Nadu using a strategic measurement setup to monitor coastal erosion has been covered in [4].

We have IoT and Machine Learning approach for prediction and monitoring of sea erosion. Which predicts with an accuracy of 98.5%.

III. PROPOSED SOLUTION

Due to the increase in problems caused by sea erosion in coastal regions, our paper provides an innovative solution through the use of IOT and machine learning technologies. Our solution involves the design of the prototype model using a rectangular box, with one side filled with sand and the other side filled with water. The entire prototype design resembles the coastal environment. We have considered temperature, wave height, wave speed, and soil moisture as the four key environmental parameters that are the main causes of sea erosion. Temperature increases lead to thermal expansion and the melting of ice, which increases the sea level. Due to an increase in the height and speed of the waves, the force at the coastline is increased, which causes erosion. Due to the increase in moisture content in the soil, it causes the land to slide into the water. For monitoring these key environmental parameters, we used four IoT sensors, which include the DHT-11 sensor for the temperature [5], the G3/4 water velocity sensor for the wave speed [6], the HC-SR04 ultrasonic sensor for the wave height [6], and the resistive soil moisture sensor for the soil moisture [7]. These sensors are deployed along the prototype design and connected to the microcontroller node MCU ESP8266 [8], which sends data to the ThinkSpeak platform. We have generated the waves in the prototype design and have collected a dataset consisting of 1,000 samples.For the machine learning component, we have used the Decision tree classification algorithm [9] due to its effectiveness in pattern recognition. After the collection of the dataset, a model is trained using the Decision tree classification algorithm on the collected dataset. Training allows the model to detect the underlying patterns and relationships among the data for an accurate prediction of sea erosion events. After training the model, the model is tested using unseen data.

After training and testing the model, it is then integrated with the web application. The user-friendly web application consists of four input fields: temperature, wave height, wave speed, and soil moisture. This allows the user to enter the values of live sensor data, and on submitting these values, the model predicts the likelihood of erosion events. In cases of sea erosion detection, stakeholders can send alerts to the local authorities or the local communities.

By combining physical prototype design, sensor deployment, Decision tree classification algorithm-based ML, and web application, our proposed solution aims to address coastal erosion problems while also promoting proactive and sustainable coastal protection.

IV. SYSTEM ARCHITECTURE

4.1 Prototype design and sensor Deployment

A prototype is designed that resembles the coastal area with sand and water. All the sensors will be installed on this to track environmental factors that cause coastal erosion. In order, to gather data, four different types of sensors are used. DHT11 for temperature, HC-SR04 ultrasonic sensor for wave height, G3/4 Water Velocity Sensor for wave height, and FC-28 resistive soil moisture sensor for soil moisture. All the sensors are connected to microcontroller devices such as Nodemcu, Arduino, etc. to enable smooth data transmission and communication.

4.2 Data collection and pre-processing

After the deployment of the sensors along the prototype, the next step is to generate waves and collect the data from the sensors. The gathered data is formed into a dataset, and it is then pre-processed. The raw dataset undergoes preprocessing such as cleaning, normalizing, and filtering. The preprocesses dataset is the basis for the ML algorithm, which is used for predicting erosion events based on previous data and patterns. The dataset is divided into a testing dataset and a training dataset.

4.3 Machine learning

After the preprocessing of the dataset, the ML model is trained and tested on the training dataset and the testing dataset, respectively. The dataset obtained is a labeled and binary classification dataset; the output is either 0 or 1, where 0 means "erosion not detected" and 1 means "sea erosion detected." For the labeled dataset, we can use supervised learning algorithms like linear regression, logistic regression, KNN classification algorithm, and Decision tree classification

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:05:32 UTC from IEEE Xplore. Restrictions apply.

algorithms.After training the model on the training dataset, the model is then tested on the testing dataset to assess how well the model generalizes to the unseen data. Among all the algorithms, the Decision tree classification algorithm has the best accuracy, which is used for urther predictions.

4.4 Web application and machine learning integration

A user-friendly web interface has been designed that is integrated with a trained and tested ML model. The user interface consists of four fields. The user enters the values from the live sensor readings, and upon submission, it displays “sea erosion detected” or “sea erosion not detected." In cases where erosion is detected, the user can send an alert to the local communities or local authorities for early action.

V. METHDOLOGIES

5.1. Sensor Integration

5.1a NodeMCU ESP8226 : NodeMCU ESP8226 is a low-cost open-source IoT platform connected with ESP8266 WiFi module. It combines the capabilities of the ESP8266 WiFi module with a development board, making it easier to program and use in IoT (Internet of Things) projects.The NodeMCU ESP8266 is an IoT development platform featuring the ESP8266 WiFi module with GPIO pins. Through the wifi module, it can connect to the internet, read the data from the sensors connected to Nodemcu, and transmits the sensor data to cloud platforms or servers.



Fig. 1 NodeMCU ESP8266

5.1b Temperature Sensor: Temperature is one of the main factors that contributes to coastal erosion. A raise in the temperature causes thermal expansion and the melting of ice, which causes raise in sea level.

In our project, we have used the DHT11 sensor to monitor the temperature. The DHT11 is an inexpensive digital temperature and humidity sensor commonly used in IoT projects. DHT11 provides accurate temperature readings with a range of 0 to 50 degrees Celsius.

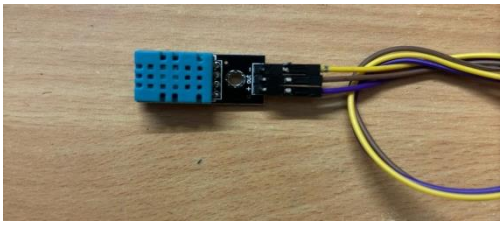


Fig. 2 DHT11 Temperature and Humidity Sensor

5.1c HC-SR04 Ultrasonic Sensor: The height of the wave is also the one of main factors contributing to coastal erosion. During the storms the height of the waves increases. The taller the waves, greater their energy. These waves hit the coastline with more force, thus resulting in the loss of more soil along the coastline. A Rise in the sea levels also increases the height of the waves. So, monitoring the height of the waves is important for the early identification of coastal erosion events. To monitor wave height, we have used the Ultrasonic sensor. Ultrasonic sensors work on the principle of time-of-flight (TOF) to measure distance. An ultrasonic wave is sent by ultrasonic sensor towards the water surface, when these ultrasonic waves hit the surface, of the water they reflect back It measures the time taken for the ultrasonic wave to travel to the water surface and reflect. By measuring the time, it calculates the distance. The change of the distance with respect to time corresponds to the change in the height of water.

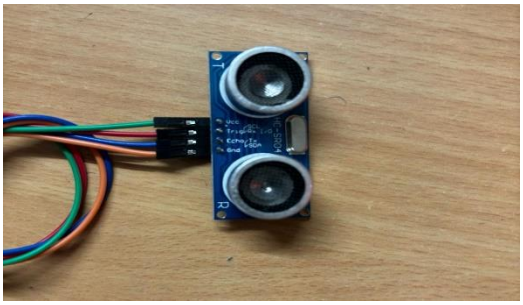


Fig. 3 HC-SR04 Ultrasonic sensor

5.1d G3/4 Water Velocity Sensor: The speed of the wave is also one among the main factors in coastal erosion. As the speed of the wave increases, it results in the increase of the wave energy. When these waves with high wave energy hit the coastline, they exert more force, resulting in soil erosion or sediment removal. Fast-Moving waves pick up and carry large amounts of sediment particles. When these waves carry more sediment particles, that results in soil erosion along the coastline. So monitoring of wave speed is crucial for early detection of erosion events. We have used the G3/4 water velocity sensor for monitoring the wave.

For measuring the wave velocity, we have used the waterflow sensor.

The formula to calculate water velocity is:

Velocity(v) =Flow Rate(Q)/Cross-sectional Area(A)

Where:

Velocity(v): Velocity(v) is the water velocity in meters per second (m/s). Flow Rate(Q): Flow Rate(Q) is the rate at which water passes through the sensor in cubic meters per second (m^3/s) or another appropriate unit per time. Cross-sectional Area(A): Cross-sectional Area(A) is the area of the pipe or channel through which the water is flowing in square meters (m^2).

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:05:32 UTC from IEEE Xplore. Restrictions apply.



Fig. 4 Water Velocity sensor

5.1e Soil Moisture Sensor: Soil moisture is also one among the main factors in coastal erosion. During the heavy rains, the moisture in the soil increases, and this excess water content in the soil results in the washing of the soil towards the coast. This can increase coastal erosion. When moisture in the soil is very high, it can cause big chunks of land to slide down into the water. So, monitoring of soil moisture is crucial for early detection of the erosion events. So, we have used the FC-28 resistive soil moisture sensor for observing the moisture in the soil. The FC-28 is a resistive soil moisture sensor commonly used in agriculture. It measures soil moisture by determining the resistance between the two implanted electrodes in the soil. All these sensors are now connected to the microcontroller Nodemcu esp8266.

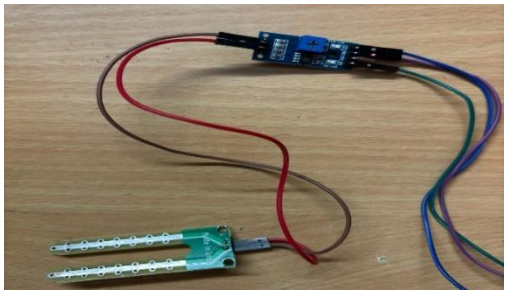


Fig. 5 FC-28 Resistive soil moisture sensor

5.2. Prototype design

Designing the prototype is an important part of our project. We have created a model that looks like a small version of a beach. We used different materials to make it look and feel like a real beach. To design a prototype, we used a rectangular tub. We placed beach soil at one end of the box and added water on the other side of the rectangular box. After developing the prototype, we have placed all the sensors: one to measure temperature, one for the speed of waves, another for the height of waves, and one to check how wet the sand is. These sensors are connected to the nodemcu, which sends the data to the Thinks peak platform.

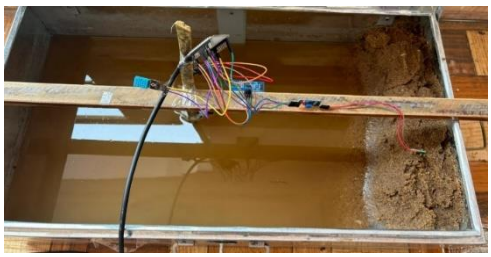


Fig. 6.. Prototype Design of Beach

5.3. Data collection and Preprocessing

After the prototype design and sensor placement, we generate the waves by varying the speeds, changing the height of the water, and varying the moisture levels and temperatures. These sensor-detected values are seamlessly transmitted by Node MCU to ThingSpeak, providing a comprehensive platform for real-time visualization and analysis of the generated wave data. We have collected a dataset of 1000 samples with four fields: wave speed, wave height, temperature, and moisture of soil. We have added an "output" field with values of 0 (no erosion) or 1 (erosion happening) based on continuous observation. We checked for and addressed any duplicate data records to eliminate redundancy and maintain dataset integrity. Examined for outliers in key fields (wave speed, wave height, temperature, and soil moisture) and implemented strategies to correct or remove them. Identified and removed any noise in the dataset, ensuring that the data accurately represents the underlying patterns. We extracted the main features from the dataset, focusing on the key variables that contribute to soil erosion prediction. After the preprocessing of the dataset, we have divided it into training data (80%) and testing data (20%). This allows for model training on a larger portion of the data and independent evaluation on a separate set.

5.4 Model Training

After data pre-processing, a machine learning algorithm can be used to train the model. We have divided the dataset into 80% for training the model and 20% for testing the model. We have used the Decision tree classification algorithm since it suits binary classification tasks, where the goal is to predict whether an instance belongs to one of two classes. Decision tree classification algorithm is a versatile algorithm that can be applied to various types of classification problems, including binary classification. In our case, the prediction is likely centred around whether sea erosion is occurring or not, making it a binary outcome.

- Decision trees represent a series of binary decisions that result in predicted outcome. It is easy to understand because it adheres to the same procedure that people use to make decisions in the actual world.
- The relationships between features and target variables are non-linear and it could be a model-complex. Decision trees are not like linear models, they can capture interactions and non-linear patterns within the data without requiring explicit feature transformations.
- Decision trees inherently perform feature selection by identifying the most informative features for splitting the data. Features that appear higher in the tree that means, closer to the root are more important in making predictions, whereas irrelevant or less important features might not be used at all.
- Decision trees are capable of handling both numerical and categorical data and requires less data cleaning relative to other algorithms. They can partition the data more effectively depending on several different types of features, making them versatile for various kinds of datasets.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:05:32 UTC from IEEE Xplore. Restrictions apply.

5.5 Model Testing

Once the model is trained successfully, the next step is to test the model on the test dataset and check for accuracy. We have used a confusion matrix, and accuracy metrics check the accuracy levels. The confusion matrix provides details about the accuracy scores. It provides true negatives, true positives, false negatives, and false positives.Our model gives 53 true positive values, 141 true negative values, 6 false negatives, and 0 false positive value. Our model trained using the Decision tree classification algorithm has an accuracy of 97%.

5.6 User Interface and alert notification

Developed a user-friendly web application for real-time sea erosion prediction using data collected from IoT sensors and ML models. The web application encompasses three main functionalities: login, prediction. We have used the HTML, CSS, JAVASCRIT, FLASK for developing the web interface and used MONGODB for the database. Developed a secure user authentication system requiring a username and password for website access. Integrated the website with the trained ML model to enable real-time predictions based on user-submitted sensor readings.

Users are redirected to an interface, where the user can enter sensor readings after logging in. The interface allows users to enter the current sensor readings (wave speed, wave height, temperature, and soil moisture) from the Thinkspeak platform. The system processes the input data using a machine learning model to predict whether erosion is occurring or not. Two possible outcomes are provided: "Erosion is Occurring," with a danger warning appearing if the prediction indicates erosion. No problem message is displayed in the case of no erosion occurrence.

If a dangerous message appears on screen, we have established a secure integration with higher authorities' contact information, ensuring that SMS alerts are sent directly to the designated recipients or local communities. We have used TWILIO API for the sms alert. The SMS alert includes a concise and informative message specifying that the system predicts. Users with administrative or higher-level access can configure SMS alert settings.

VI . RESULTS

When we generate the waves along the prototype design, the sensors placed along the prototype design read the values and send the readings to the ThinkSpeak cloud platform. The data is represented in terms of graphs with respect to the time and the individual sensor readings.

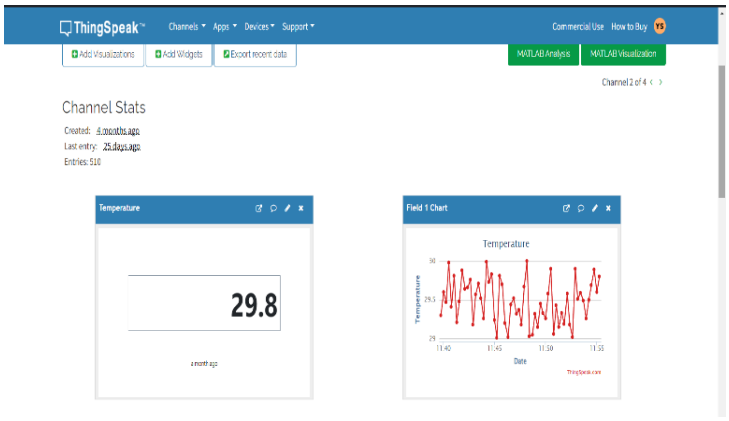


Fig. 7 Temperature readings



Fig. 8 Wave speed readings



Fig. 9 Wave height readings

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:05:32 UTC from IEEE Xplore. Restrictions apply.

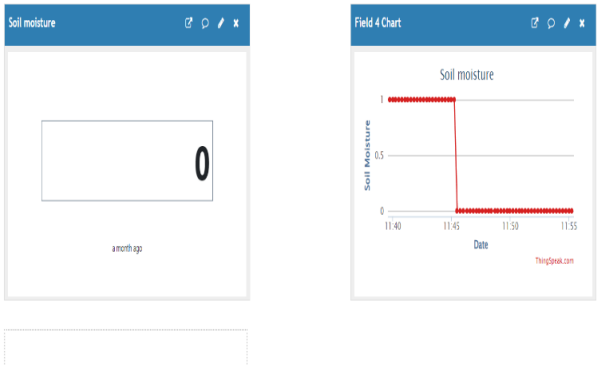


Fig. 10 Soil Moisture readings

These readings are accessed by the user or higher authorities for the prediction and monitoring of sea erosion events. For the prediction of the erosion, the authorities have access to the user interface.

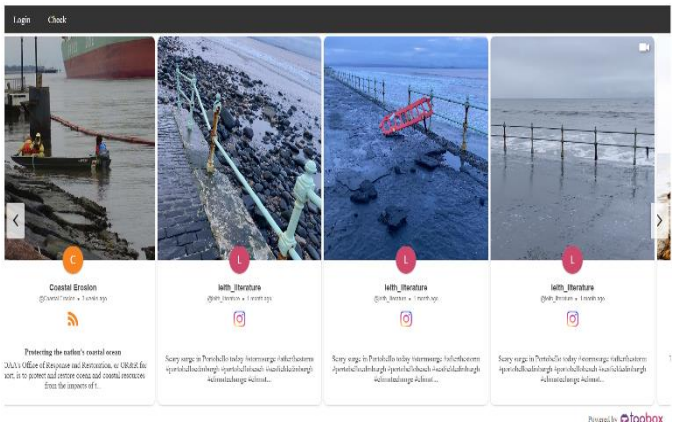


Fig. 11 Main page of user interface

This is the initial page when the authorities open the website, where they login for the prediction of the erosion events and check the previous erosion events by clicking the “Click” option.

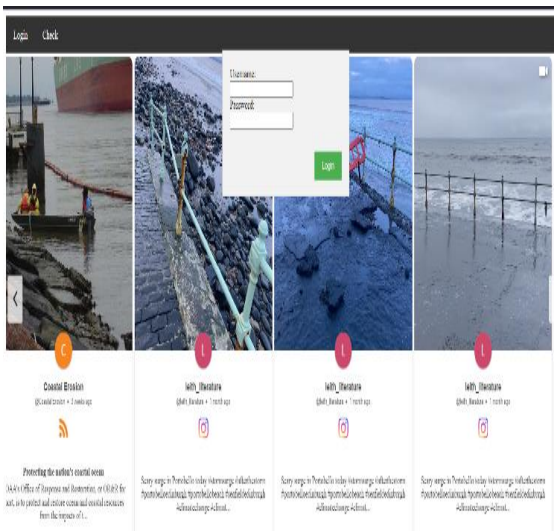


Fig. 12 Login page

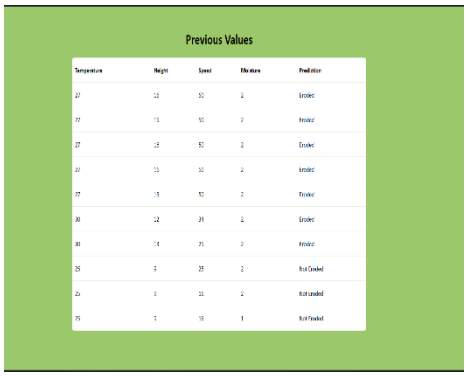


Fig. 13 previous erosion events web page

After logging in, the user will be directed to a web page with four input fields: temperature, wave height, wave speed, soil moisture, and a submit button.

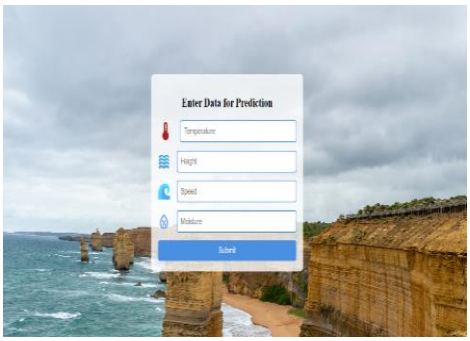


Fig. 14 Web page for entering the sensor readings

When the user enters the values of temperature, wave height, wave speed, and soil moisture and clicks the submit button, if an erosion event is detected, they will be redirected to another page; otherwise, it displays “Sea erosion not detected”.



Fig.15 webpage when erosion detected

If there are continuous soil erosion occuring, the user can click on the “Send SMS Alert” button and send an alert to local communities and authorities for early prevention of damage.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:05:32 UTC from IEEE Xplore. Restrictions apply.

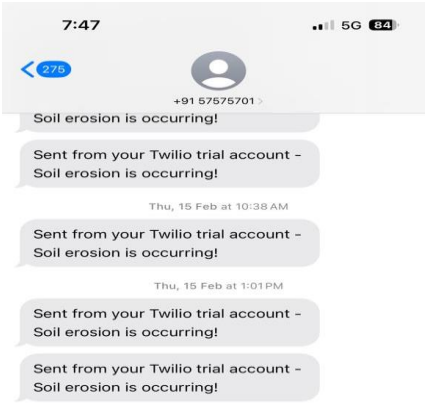


Fig. 17 SMS alert

VII. CONCLUSION

The sea erosion monitoring and prediction project integrates four types of sensors, which includes a soil moisture sensor, a temperature sensor, an ultrasonic sensor for height measurement, and a water flow sensor for wave velocity. All these sensors are linked to the NodeMCU microcontroller. This entire sensor system allows for monitoring of environmental conditions, providing valuable details and factors that contribute to coastal erosion. These sensors are deployed along the prototype design, and NodeMCU sends the sensor readings to the ThinkSpeak platform.

By using the Decision tree classification machine learning algorithm to train the model, it gives results with an accuracy of 97% ,and that is used for further predictions.

The developed user interface acts as a bridge between users and the machine learning model trained and tested using Decision tree classification algorithm. When the model predicts an erosion event based on sensor readings, the alert system can be triggered by stakeholders or authorities to local authorities or local communities, enabling timely intervention to mitigate or prevent potential damage.

This entire approach not only provides a detailed view of the current environmental conditions but also establishes a foundation for future research and data-driven decision-making in coastal management and erosion prevention. The project shows the importance of combining IoT technology and machine learning technologies to monitor and safeguard sea reas, contributing to early prediction of erosion events in coastal areas and early prevention of the damage caused by the erosion.

REFERENCES

[1]. Zhang, K., Douglas, B.C. & Leatherman, S.P. Global Warming and Coastal Erosion. *Climatic Change* **64**, 41–58 (2004). https://doi.org/10.1023/B:CLIM.0000024690.32682.48

[2]. Roberto Girau, Member, IEEE, Matteo Anedda, Member, IEEE, Mauro Fadda, Member, IEEE, Massimo Farina,Alessandro Floris, Member, IEEE, Mariella Sole, Daniele Giusto, Senior Member, IEEE "Coastal Monitoring System based on Social Internet of Things Platform" DOI: 10.1109/JIOT.2019.2954202.

[3]. Papakonstantinou, M. Doukari, and K. Topouzelis, "Coastline changedetection using unmanned aerial vehicles and image processing," Fre-senius Environmental Bulletin, vol. 26, pp. 5564–5571, Sept. 2017

[4]. Jena BK, Sathish Kumar D, Kintada K. Operational strategy to monitor coastal erosion in tropical areas. The International Journal of Ocean and Climate Systems. 2017;8(3):135-143. doi:10.1177/1759313117704837

[5]. Rizki juli syahputra, "Monitoring The Temperature And Humidity Air In The Room Using A Sensor IOT-Based DHT-11", *j. of artif. intell. and eng. appl.*, vol. 3, no. 1, pp. 363–367, Oct. 2023.

[6]. T H Nasution et al 2018 IOP Conf. Ser.: Mater. Sci. Eng. 308 012031 arXiv:2210.03019 [cs.NI] "Comparative Analysis and Calibration of Low Cost Resistive and Capacitive Soil Moisture Sensor" DOI: https://doi.org/10.48550/arXiv.2210.03019

[7]. "Internet of Things and Nodemcu A review of use of Nodemcu ESP8266 in IoT products", International Journal of Emerging Technologies and Innovative Research (www.jetir.org | UGC and issn Approved), ISSN:2349-5162, Vol.6, Issue 6, page no. pp1085-1088, June 2019, A

[8]. Harsh H. Patel, Purvi Prajapati, "Study and Analysis of Decision Tree Based Classification Algorithms," *International Journal of Computer Sciences and Engineering*, Vol.6, Issue.10, pp.74-78, 2018.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 12,2026 at 09:05:32 UTC from IEEE Xplore. Restrictions apply.